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Two-phase multi-objective optimization of blood supply chains with mobile donation sites and perishability ☆

Vincent F. Yu [a], Annisa Cipta Nabila [a,c], Pham Tuan Anh [b], Soumen Kumar Das [a,*], Achmad Pratama Rifai [c]

[a] Department of Industrial Management, National Taiwan University of Science and Technology, Taipei, Taiwan
[b] College of Technology and Design, University of Economics Ho Chi Minh City, Viet Nam
[c] Department of Mechanical and Industrial Engineering, Universitas Gadjah Mada, Daerah Istimewa Yogyakarta, Indonesia

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ABSTRACT

Designing an efficient blood supply chain (BSC) network requires careful consideration of the perishable nature of blood products and variations in supply and demand over multiple periods. This study develops a two-phase, multi-objective optimization framework for BSC network design to enhance accessibility and responsiveness while minimizing total network cost and total flow time. Phase I generates candidate mobile donation sites using k-means clustering, while Phase II formulates a math model to optimize collection, production, inventory, and distribution decisions. The model considers multiple blood products with different shelf lives, transportation modes with varying capacities and speeds, and inventory decisions under perishability constraints. The ϵ-constraint method is applied to analyze the trade-off between economic and social objectives. Numerical experiments based on real-world data from an Indonesian BSC show that dynamic, period-varying mobile donation sites improve responsiveness with negligible additional cost while requiring fewer opened sites than a fully static design. The results also indicate that the model can reduce shortages and outdated inventory under changing supply and demand conditions. Under progressive supply reduction, the social objective avoids shortages entirely by activating mobile collection. A network analysis further shows that raw-material supply, rather than processing capacity, is the binding resource governing performance. Overall, the proposed model provides an integrated framework for supporting BSC network design in multi-period and multi-product settings.

1. Introduction

Blood is a perishable, irreplaceable resource that cannot be synthesized and can only be obtained through voluntary donation, which makes the design of an efficient blood supply chain (BSC) a long-standing concern in the operations research literature (Prastacos, 1984). This results in highly volatile supply and demand patterns across periods, creating significant challenges for effective management. Additionally, the perishable nature of blood imposes strict limits on storage, as large quantities cannot be stored for long periods without compromising safety and usability. These factors highlight the need for adaptive strategies to ensure the reliability and effectiveness of the BSC network.

A BSC network is typically organized around components, including collection, production, inventory, and distribution (Osorio et al., 2015), each of which imposes its own operational constraints. At the collection stage, blood is gathered through fixed donation centers and temporary or mobile sites (Arani et al., 2021; Eskandari-Khanghahi et al., 2018; Seyfi-Shishavan et al., 2023). At the production stage, whole blood is fractionated into components such as red blood cells, plasma, and platelets (Hosseini et al., 2023; Osorio et al., 2017), each product type with a specific shelf life and therefore a different risk of wastage, with short-lived components such as platelets receiving particular attention (Eskandari-Khanghahi et al., 2018; Suen et al., 2023). The distribution

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* Corresponding author.
E-mail addresses: vincent@mail.ntust.edu.tw (V.F. Yu), annisaciptanabila@mail.ugm.ac.id (A.C. Nabila), ptanh@ueh.edu.vn (P.T. Anh), soumen@mail.ntust.edu.tw (S.K. Das), achmad.p.rifai@ugm.ac.id (A.P. Rifai).

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Table 1
Summary of previous research related to blood supply chains.

References	Objective	Problem features								Method
		Collection		Production		Inventory		Distribution		
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
Duan and Liao (2014)	Ec			✓			✓			TA-TS
Osorio et al. (2017)	Ec		✓			✓	✓			MILP
Eskandari-Khanghahi et al. (2018)	S, Ec	P			✓		✓	✓		PO
Larimi et al. (2019)	S, Ec				✓		✓			SP
Hamdan and Diabat (2020)	S, Ec	P								LR
Khalilpourazari et al. (2020)	S, Ec	P						✓		NL
Arani et al. (2021)	S, Ec	P		✓		✓		✓		GP
Dehghani et al. (2021)	Ec	P		✓			✓		✓	SP
Shokouhifar et al. (2021)	Ec						✓		✓	WO
Hosseini et al. (2023)	S, Ec	P				✓	✓	✓		NSGA-II
Mansur et al. (2023)	Ec	P				✓				MILP
Khalilpourazari and Hashemi Doulabi (2023)	S, Ec	P				✓	✓			RO
Momenitabar et al. (2023)	S, Ec			✓			✓			Fuzzy-MILP
Suen et al. (2023)	S, Ec						✓			GP
Seyfi-Shishavan et al. (2023)	Ec	P								FO
Tirkolaee et al. (2023)	S, Ec	P						✓		PO
Ala et al. (2024)	S, Ec	P			✓					RO
Entezari et al. (2024)	S, Ec	P		✓		✓				Augmecon
Rekabi et al. (2024)	S, Ec					✓	✓	✓		LR
Abdolazimi et al. (2025)	S, Ec	P				✓				GA
This Research	**S, Ec**	C	✓			✓	✓	✓		ϵ-constraint

Notes: **Features:** (1) Facility location; (2) Collection bag type; (3) ABO-Rh blood type; (4) Laboratory facility; (5) Multiple products; (6) Perishability; (7) Multiple vehicles; and (8) Transshipment. Objectives: Ec: Economic; S: Social; P: Predetermined; C: Clustering. **Methods:** Augmecon: Augmented ϵ-constraint; FO: Fuzzy optimization; GA: Genetic algorithm; GP: Goal programming; LR: Lagrangian relaxation; NL: Neural learning; NSGA-II: Non-dominated sorting genetic algorithm II; PO: Possibility optimization; RO: Robust optimization; SP: Stochastic programming; TA-TS: Threshold accepting-Tabu search; WO: Whale optimization; MILP: Mixed-Integer Linear Programming.

stage presents additional logistical challenges, including the selection of appropriate vehicles with varying capacities (Tirkolaee et al., 2023) and incorporating transshipment between inventory sites (Larimi et al., 2019). Effective vehicle selection plays a crucial role in the allocation and delivery of blood products, ensuring timely distribution to meet fluctuating demand (Hosseini et al., 2023). Demographic and geographic factors further complicate the blood supply chain and often create barriers to the efficient distribution of blood products (Gunpinar & Centeno, 2016).

From an economic standpoint, network design must minimize the cost of establishing and operating collection, production, inventory, and distribution activities (Rekabi et al., 2024; Zahiri & Pishvaee, 2017). From a social standpoint, because blood products serve a healthcare need rather than a discretionary demand, outcomes such as unmet demand and outdated inventory carry consequences beyond cost, and responsiveness becomes critical given that many transfusions occur under emergency conditions where delay has direct clinical consequences (Hosseini et al., 2023). Mobile donation sites have received particular attention in the BSC literature, yet their locations are typically optimized once and then treated as fixed or predetermined inputs (Hamdan & Diabat, 2020; Hosseini et al., 2023; Jabbarzadeh et al., 2014), limiting the network's ability to adapt to shifting donor availability and demographic conditions over time.

Research on BSC network design has addressed these challenges in several directions. Multi-period and multi-product features have been investigated to capture the time-varying and perishable nature of blood products (Eskandari-Khanghahi et al., 2018; Hosseini et al., 2023), while multi-objective methods have been used to reconcile economic and social goals (Rekabi et al., 2024). However, each of these elements, formulations of the BSC, perishability constraints, multi-period planning, and ϵ-constraint optimization, is already well established in the literature. What has received comparatively little attention is whether collection-site locations can be determined dynamically, rather than fixed in advance, without compromising the tractability of the resulting multi-echelon, multi-product optimization model.

Motivated by these gaps, this study addresses four research questions: (i) how can a BSC network jointly coordinate collection, production, inventory, and delivery decisions while accounting for multi-product perishability and heterogeneous vehicles; (ii) how can dynamic, period-varying mobile donation sites be generated and integrated into multi-period operational planning; (iii) how does dynamic mobile-site selection affect responsiveness and site deployment compared with fixed-location and no-mobile-site designs; and (iv) how can the trade-off between economic cost and responsiveness be analyzed using the ϵ-constraint method? The main contributions of this study are as follows.

- We develop a comprehensive BSC network model that spans all four echelons, collection, production, inventory, and delivery, incorporating multi-product perishability, system time, and vehicle heterogeneity within a single integrated formulation.
- We propose a two-phase optimization framework that separates mobile-site candidate generation from multi-period operational planning. Phase I uses clustering to identify dynamic, period-varying mobile donation sites, while Phase II solves a multi-period, multi-product MILP over the resulting network. This structure keeps the operational model linear and tractable while allowing mobile donation sites to adapt across periods.
- We consider outdating and disposal as operational penalty costs within the production and inventory stages, so that the model reduces waste as a cost-efficiency mechanism, and we quantify the resulting waste behavior across demand-increase and supply-reduction scenarios.
- We apply the ϵ-constraint method to jointly analyze economic and social trade-offs, using a turning-point analysis to identify where the two objectives balance, rather than treating either objective in isolation as in most prior BSC studies.

The methodological contribution of this study is therefore not the individual components, multi-period planning, perishability, or ϵ-constraint optimization, each of which has precedent in the literature, but their integration through a two-phase structure that resolves a combinatorial, nonlinear location problem separately from a linear operational problem, together with the extension of dynamic (rather

than predetermined) mobile-site selection to a multi-product, multi-echelon BSC setting, a combination not previously addressed in the BSC literature summarized in Table 1.

The rest of the paper is structured as follows. Section 2 conducts a literature review of BSC and highlights the research gaps addressed in our proposed problem. The development of the mathematical model for the proposed network is presented in Section 3. Section 4 explains the solution methodology utilized in this research. The results of the numerical experiments are presented in Section 5. Section 6 provides further insights through a real-world numerical case. Finally, directions for future research are discussed in Section 7.

2. Related works

This section reviews the main streams of literature related to the present study. It discusses research on the blood supply chain, perishable products, multi-periodicity, and multi-objective optimization.

2.1. Blood supply chain

The blood supply chain generally manages the flow of blood products from donors to patients through several echelons (Pirabán et al., 2019). Research on BSC network design has expanded significantly over the years and has addressed increasingly complex challenges. Meneses et al. (2023) and Osorio et al. (2015) described the BSC network in terms of four main components: collection, production, storage, and distribution. Meneses et al. (2023) further highlighted major directions and key issues in BSC modeling and noted that strategic, tactical, and operational decisions should be considered in a multi-period context.

Prior research on the BSC has addressed a wide range of planning and operational challenges. Among these, many studies have focused on demand and supply conditions, which are often addressed through forecasting techniques (Fanoodi et al., 2019; Shokouhifar & Ranjbarimesan, 2022). Optimization approaches have also been widely applied in BSC management. For example, robust optimization has been used to develop more resilient planning models (Ramezanian & Behboodi, 2017). Recent studies have further highlighted the importance of uncertainty and sustainability in BSC modeling. Sheibani et al. (2025) incorporated environmental and social considerations into a multi-objective BSC network design under uncertainty, while Torrado and Barbosa-Póvoa (2022) reviewed the main directions in sustainable BSC network design under uncertain conditions. In addition, BSC design has been studied in disaster response context (Samani et al., 2018), under disruptive conditions (Entezari et al., 2024), and during the COVID-19 pandemic (Abdolazimi et al., 2025; Hosseini et al., 2023; Mondal & Roy, 2021).

Designing an effective BSC requires careful attention to decision-making processes, objectives, and operational factors such as donor scheduling and collection planning. Previous studies have also addressed routing optimization at the collection stage to ensure timely and efficient blood transportation (Gunpinar & Centeno, 2016). At the tactical level, much of the literature focuses on inventory management under perishability and fluctuating demand, while Osorio et al. (2017) discussed production planning in the blood supply chain. At the strategic level, earlier studies examined facility location decisions and capacity planning. Because blood is highly perishable, strict constraints apply across all planning levels, requiring rapid processing and distribution as well as close coordination throughout the entire supply chain. Although this body of work spans all four echelons, most network models still consider collection-site locations as predetermined inputs; the adaptive, demand-responsive placement of mobile donation sites has received comparatively little attention in recent BSC optimization studies, which is a central motivation for the present work.

2.2. Perishable product

Perishable products are items with limited shelf lives that are prone to spoilage or quality loss over time (Shekarabi & Mozdgir, 2025). These products are closely associated with securability, which refers to the ability to preserve quality, and spoilability, which refers to the tendency of a product to deteriorate (Kashyap et al., 2026). Their natural deterioration requires specific environmental conditions, such as temperature control, humidity regulation, and hygiene standards, to maintain usability (Shekarabi & Mozdgir, 2025).

In inventory management, perishable items require a delicate balance between minimizing shortages and avoiding excess stock that may expire before use (Dehghani et al., 2021). Effective handling is essential due to the time-sensitive nature of these products. Biza et al. (2024) emphasized the increasing need to enhance perishable product management, with most studies centered on food-related applications. Wu and Tanaka (2025), for instance, proposed an optimization-based inventory control approach that integrates perishability and multiple cost considerations.

Perishability is also strongly linked to inventory policy. Gulecyuz et al. (2025) compared FIFO (First-In, First-Out) and LIFO (Last-In, First-Out) policies, particularly in the food sector. In healthcare, Katsaliaki et al. (2014) identified blood as a key example of a perishable product and used game theory to examine supplier and distributor strategies in the blood supply chain. The shelf life of blood products varies significantly. Red blood cells can last up to 42 days, frozen plasma up to one year, and platelets only about 5 days. This makes timely inventory turnover especially important.

In the production echelon of the BSC, complexity increases because of the fractionation process, which separates whole blood into components such as red blood cells, plasma, and platelets (Osorio et al., 2015). This creates a multi-product setting with different handling and storage requirements. Although some studies have considered multi-product settings, limited attention has been given to the collection stage, where variations in blood bag types and donation methods, such as whole blood and apheresis, affect subsequent processing and logistics (Shokouhifar et al., 2021). Furthermore, strategic decisions like laboratory location and production reliability remain underexplored, despite their impact on processing time and potential waste (Hosseini et al., 2023).

Perishability is also a core issue in the inventory echelon, particularly for short-lived components like thrombocyte concentrate (TC) (Larimi et al., 2019; Shokouhifar et al., 2021; Suen et al., 2023). While FIFO and LIFO policies have been applied in related studies (Zhou et al., 2021), only limited attention has been given to inventories involving multiple blood products with different shelf lives. In addition, inventory-related challenges in the BSC often rely on simulation-based approaches, which may not fully capture the complexities of optimization-based decision-making (Ekaputri & Kusumastuti, 2022).

In parallel, queue management at donation centers also affects timeliness. Queuing theory has been employed to enhance donor flow efficiency, with FIFO typically preferred in healthcare settings for ethical and operational reasons (Gulecyuz et al., 2025). Therefore, both inventory and queuing decisions must be sensitive to perishability to minimize loss and maintain service quality. In response to these challenges, this study develops a strategic-level, multi-period model that incorporates the perishability of blood products and supports timely, cost-efficient, and coordinated decision-making across the BSC network.

2.3. Multi-periodicity

Multi-period modeling plays an important role in supporting effective decision-making in supply chain management. Li and Cruz (2022) demonstrated that using a multi-period planning horizon allows decision-makers to evaluate the long-term effects of their actions and

identify changing patterns within the supply network. This time-based perspective is particularly important in contexts where demand, supply, and operational environments are dynamic.

In line with growing sustainability concerns, consumer behavior is shifting toward greener choices, prompting decision-makers to invest in environmentally friendly technologies. As noted by Saberi et al. (2019), such investment decisions must balance environmental goals with profit maximization. This balance is better analyzed in a multi-period framework, which can capture both short-term trade-offs and long-term benefits. Multi-period optimization is also necessary in highly variable operational contexts. Yang and Wang (2025) emphasized that in systems with fluctuating supply or demand, such as healthcare logistics, a multi-period approach enables more accurate scheduling and resource allocation at the sub-period level. This is especially relevant for blood supply chains, where fluctuations in donor availability, processing times, and demand urgency present significant coordination challenges.

In the distribution echelon of the BSC, scheduling and transportation decisions must align with the perishability and urgency of blood products. Previous studies have examined multi-vehicle routing decisions to optimize delivery (Eskandari-Khanghahi et al., 2018; Rekabi et al., 2024). Transshipment between hospitals has also been proposed to improve supply flexibility and meet demand more effectively (Kees et al., 2022; Larimi et al., 2019; Zhou et al., 2021). However, limited studies have considered the impact of travel time on product viability, an essential factor for highly perishable components (Hosseini et al., 2023). In addition, Shaw et al. (2025) studied a humanitarian redistribution problem using a multi-period allocation model, which further supports the importance of time-based planning under uncertain and changing operational conditions.

Moreover, fluctuations in both supply and demand, together with the time required for processing and testing, make a multi-period perspective necessary across all four echelons. From donor scheduling and laboratory processing to distribution planning, a time-sensitive and adaptive framework is needed to ensure efficiency and minimize product waste. These interconnected challenges underscore the importance of integrating multi-period planning into BSC network design so that the system remains responsive, resilient, and able to optimize outcomes over different time horizons.

2.4. Two-phase and multi-objective optimization

In the BSC, the collection process often relies on predetermined locations or fixed facilities, such as static blood donation centers, hospitals, or scheduled mobile donation units (Ala et al., 2024; Arani et al., 2021). Similarly, the locations of laboratories and mobile blood collection sites are often decided in advance and do not adapt to changing demand or population distribution. This dependence on fixed infrastructure creates challenges, especially for geographically dispersed or demographically diverse populations that may face difficulties in accessing these sites.

To address this limitation, more flexible modeling approaches have been introduced in supply chain planning. One such approach is two-phase optimization, in which the first phase explores possible solutions and the second phase identifies the final optimal solution (Schoen, 2002). This approach has been applied in demand-responsive and supply chain network contexts to combine exploration and optimization more effectively (Huang et al., 2020; Lambert & Spliethoff, 2024).

In the blood supply chain, a notable gap remains at the collection stage, particularly in the strategic placement of temporary facilities or mobile donation sites. Within a two-phase optimization framework, the first phase can be used to explore potential locations for such temporary facilities. Among the available location selection techniques, clustering methods have proven useful, with k-means clustering being one of the most widely used methods (Apon et al., 2006). This exploratory step is especially relevant given the diverse range of objective functions used in recent BSC studies.

In the broader context of BSC modeling, multi-objective functions are commonly used to address diverse and often conflicting goals across economic, social, and environmental dimensions. Economic objectives typically aim to minimize total system costs, including facility establishment, inventory holding, transportation, and shortage costs (Rekabi et al., 2024; Zahiri & Pishvaee, 2017). In contrast, some studies pursue profit maximization, accounting for varying product values across supply chain levels (Mansur et al., 2023). Social objectives focus on increasing employment opportunities (Arani et al., 2021; Tirkolaee et al., 2023), improving blood flow and reducing donor/patient waiting times (Rekabi et al., 2024), and minimizing unmet demand Zahiri and Pishvaee (2017). Environmental objectives seek to reduce carbon emissions (Das & Roy, 2019; Rekabi et al., 2024), minimize the physical footprint of facilities (Arani et al., 2021), and reduce waste, either as an explicit goal or a penalty function (Meneses et al., 2023). Similarly, Yu et al. (2024) proposed a multi-objective two-stage approach under a carbon cap-and-trade policy, showing how economic and environmental goals can be handled within the same optimization framework.

To solve such complex problems, the ϵ-constraint method is widely adopted in BSC and broader supply chain studies. Mavrotas (2009) introduced the ϵ-constraint as an effective way to transform a multi-objective problem into a series of single-objective problems by treating the remaining objectives as constraints bounded by ϵ values. This method has been applied in biomass supply chains (Wang et al., 2024) and location-inventory-routing within circular economy frameworks (Govindan et al., 2023). Although two-phase optimization, clustering-based facility location, and the ϵ-constraint method have each been employed separately in recent studies, their integration for dynamic, period-varying mobile-site selection within a multi-product, perishable, socio-economic BSC network has not yet been addressed, which is the specific gap this study targets.

2.5. Research gap

Table 1 summarizes the most recent and relevant studies on blood supply chain network design and highlights the key differences and contributions of this study. Despite the growing use of multi-objective approaches in BSC modeling, several research gaps remain.

- In the collection echelon, many models rely on predetermined infrastructure and do not consider adaptive or demand-responsive site allocation, especially in areas with demographic and geographic limitations.
- In the production echelon, studies have not fully addressed the strategic placement of laboratories and the reliability of the blood fractionation process, particularly in different product types.
- In the inventory echelon, most models handle single-product inventory and rarely consider multi-product perishability with varying shelf lives.
- In the distribution echelon, time-sensitive decision variables such as vehicle scheduling, travel time effects, and transshipment strategies are often not well integrated.

Moreover, many existing models still use single-objective formulations or focus only on isolated components of the supply chain, such as inventory or transportation, while overlooking interactions across echelons and realistic system dynamics. Therefore, there is a clear need to develop a more integrated, multi-objective, and multi-echelon model that better reflects the complexity of BSC operations, especially in the context of perishability.

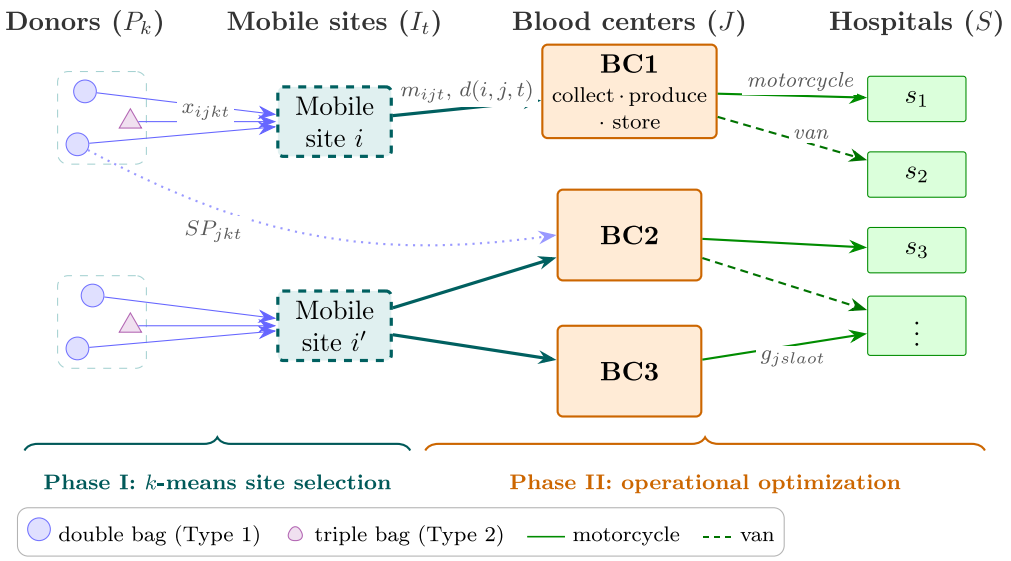


Fig. 1. Proposed blood supply chain network design.

3. Problem formulation

This section formalizes the proposed two-phase blood supply chain network design. We first define the problem and its network structure, then state the modeling assumptions and notation, and finally present the mathematical model, which comprises Phase I clustering formulation and Phase II multi-objective MILP.

3.1. Problem definition

The proposed problem is formally described as a two-phase multi-objective location–allocation network problem. The BSC network design aims to optimize both economic and social objectives by minimizing total cost and the total flow time of goods within the network, respectively.

A multi-period framework is adopted, which allows donor points to vary over time and better reflect real-world conditions. To support this flexibility, a clustering-based exploration is introduced in the first phase of the model to determine optimal mobile donation sites for each period. The proposed BSC network, displayed in Fig. 1, consists of four critical echelons: donor collection, production and inventory at blood centers (BCs), transportation, and hospital demand points. Blood is collected through both mobile donation sites, whose locations vary each day, and fixed donation sites, namely BCs, which serve as collection, production, and inventory hubs. In this model, the distances between mobile sites and blood centers, as well as between blood centers and hospitals, are explicitly considered to reflect logistical challenges and their impact on both time and cost.

In the second phase, the selected mobile donation sites from the first phase are utilized to determine production, inventory, and distribution decisions across the network. Blood collected using multi-collection bags undergoes fractionation at the production stage, as shown in Fig. 2, yielding multiple blood products with different shelf lives. After fractionation, laboratory testing is conducted to ensure product quality. Units that exceed production capacity or fail to meet quality standards are categorized as waste. Approved products are stored in inventory and issued following a first-in-first-out (FIFO) principle (Dehghani et al., 2021), consistent with clinical practice. The age-indexed inventory formulation in Section 3 tracks each unit by age so that outdating is accounted for explicitly; as verified in Section 6, because the shelf lives of the products considered exceed their typical inventory residence times, the quantity of outdated product is invariant to the issuing order. Products that exceed their maximum shelf life during storage become outdated waste (Osorio et al., 2017). This perishability constraint is critical, given the different lifespans of each blood product.

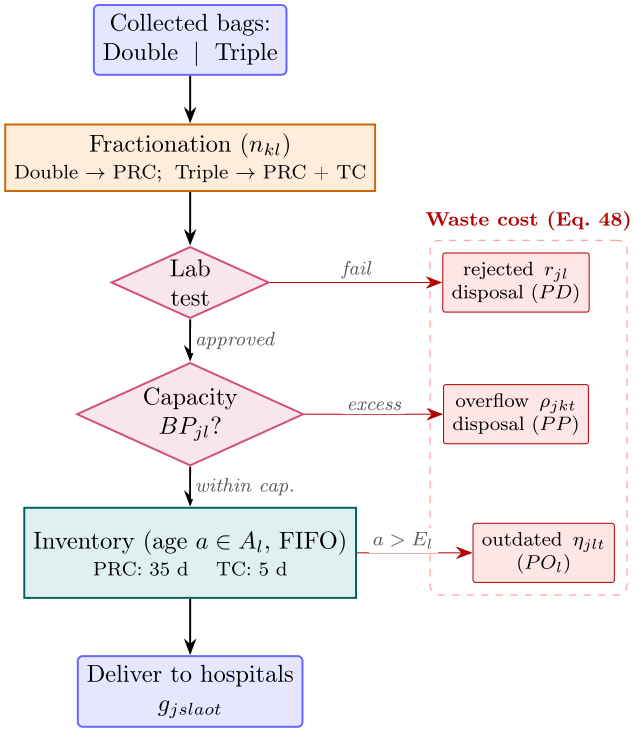


Fig. 2. Blood products production process.

In the delivery echelon, multiple vehicle types are considered, each characterized by different transportation speeds and cost structures (Tirkolaee et al., 2023). This allows the model to evaluate various trade-offs between delivery responsiveness and cost-efficiency. One of the social objectives of this study is to improve responsiveness, defined as the total time blood products spend in the system, from collection to delivery. At the same time, the economic objective is to minimize total

Table 2
Sets used in the model.

Set	Definition
I	Set of potential mobile donation sites (mobile collection points), indexed by i
J	Set of blood center sites (fixed collection points), indexed by j
T	Set of time periods in days, indexed by t
I_t	Set of mobile donation sites identified in Phase I for period $t \in T$ and supplied to Phase II as candidate locations, where $I_t \subseteq I$
K	Set of blood bag collection types, indexed by k
P	Set of all donor locations (supply points)
P_k	Subset of donor locations for bag collection type $k \in K$, indexed by p
S	Set of hospitals (demand points), indexed by s
L	Set of blood product types, indexed by l
O	Set of delivery vehicles, indexed by o
A_l	Set of ages of blood product l, indexed by a

network cost, including facility operations, transportation, inventory management, and waste handling.

The model also incorporates waste-related cost components, particularly for products discarded due to capacity overflow, failed testing, or expiry. Embedding these penalties into the cost structure drives the model toward reduced outdating and disposal as a matter of cost efficiency. Overall, the proposed model offers a realistic and comprehensive framework for designing a multi-period, multi-product, and multi-objective blood supply chain, capturing the trade-offs between cost-efficiency and service responsiveness.

3.2. Assumptions and notation

To develop the mathematical model, assumptions are introduced. This section also presents the notation used in the mathematical formulation.

The proposed model is based on the following assumptions.

- Blood supply and hospital demand are assumed to be deterministic in each period.
- The model assumes that both donor availability (supply) and hospital needs (demand) are known with certainty in each period.
- There are two types of collection sites: mobile donation sites and fixed blood centers. Mobile site locations may vary by period, while fixed centers remain constant.
- Donors are grouped according to the type of blood bag used, which reflects the collection method and affects the resulting blood products.
- All inter-location distances are modeled using Euclidean distance.
- The failure rate in laboratory testing is known and fixed for each blood product type.
- Shortages are allowed, but each unit of unmet demand incurs both a penalty cost and a penalty time, representing the economic and service-level consequences.
- Blood product transfers between blood centers are not allowed.

The notation used in the model, including sets, parameters, and decision variables, is presented below in Tables 2–4. In addition, for location descriptions, we use superscripts (x) and (y) to indicate the longitude and latitude coordinates of the nodes.

3.3. Mathematical model

To manage the variation of supply points across different periods, we formulate a two-phase modeling framework.

3.3.1. Phase I: clustering formulation

Phase I determines the locations of mobile donation sites for each period by assigning supply points to collection sites and computing cluster centroids. The formulation follows a k-means logic augmented with coverage constraints, and is guided by the objective (3.1), which minimizes the number of activated mobile sites.

$$\min F_1 = \sum_{t\in T}\sum_{i\in I} u_{it}. \tag{3.1}$$

Subject to:

$$\sum_{i\in I_t} u_{it} \le \sum_{j\in J} NM_j, \qquad \forall t \in T, \tag{3.2}$$

$$\sum_{i\in I_t} w_{ipkt} + \sum_{j\in J} w_{jpkt} = 1, \qquad \forall p \in P_k, k \in K, t \in T, \tag{3.3}$$

$$\sqrt{(i_t^x - j^x)^2 + (i_t^y - j^y)^2} \le R_{ij}, \qquad \forall i \in I_t, t \in T,\ \exists j \in J, \tag{3.4}$$

$$w_{jpkt}\sqrt{(j^x - p_{kt}^x)^2 + (j^y - p_{kt}^y)^2} \le R_j, \qquad \forall j \in J, p \in P_k, k \in K, t \in T, \tag{3.5}$$

$$i_t^x = \frac{1}{\sum_{k\in K}\sum_{p\in P_k} w_{ipkt}} \cdot \sum_{k\in K}\sum_{p\in P_k} p_{kt}^x \cdot w_{ipkt}, \qquad \forall i \in I, t \in T, \tag{3.6}$$

$$i_t^y = \frac{1}{\sum_{k\in K}\sum_{p\in P_k} w_{ipkt}} \cdot \sum_{k\in K}\sum_{p\in P_k} p_{kt}^y \cdot w_{ipkt}, \qquad \forall i \in I, t \in T, \tag{3.7}$$

$$\sum_{k\in K}\sum_{p\in P_k} w_{ipkt} \le \sum_{k\in K} P_{kt} \cdot u_{it}, \qquad \forall i \in I, t \in T. \tag{3.8}$$

Constraint (3.2) limits the maximum number of mobile facilities that can be deployed in each period. Constraint (3.3) ensures that each supply point is covered by exactly one collection site, either a mobile donation site or a blood center. Since the coverage radius of both facility types is known, constraint (3.4) ensures that the cluster location lies within the desired coverage radius of at least one blood center. Constraint (3.5) defines the coverage of blood center collection sites. In both constraints, the Euclidean distance is expressed in the same distance units as the corresponding radius; in Constraint (3.5), the coverage condition is activated when $w_{jpkt} = 1$. Constraints (3.6) and (3.7) determine the centroid coordinates of each cluster in each period. Finally, Constraint (3.8) is the cluster activation constraint, which ensures that a potential mobile donation site i in period t is activated only when at least one supply point is assigned to it. The first phase produces a set of selected mobile donation site locations for each period, along with the number of available donors (supply) at each selected mobile site and at the fixed donation sites, represented by blood centers j. The results obtained from the first phase in each period are then used as input parameters for the second phase. The following equation describes the linkage between the first and second phases.

$$\sum_{p\in P_k} w_{ipkt} = SP_{ikt}, \qquad \forall i \in I_t, k \in K, t \in T. \tag{3.9}$$

$$\sum_{p\in P_k} w_{jpkt} = SP_{jkt}, \qquad \forall j \in J, k \in K, t \in T, \tag{3.10}$$

$$u_{it} = 1, \qquad \forall i \in I_t, t \in T. \tag{3.11}$$

3.3.2. Phase II: MILP formulation

In the second phase, the feasible set of mobile locations for each period, as defined in Eq. (3.11), alongside the available supply quantities

Table 3
Parameters used in the model.

Parameter	Definition
NM_j	Maximum number of mobile collection vehicles deployable from blood center $j \in J$
FM	Fixed cost of mobile donation site establishment
CC_k	Unit collection cost for blood bag type k
CT	Transportation cost from a mobile site to a blood center
CP_l	Unit production cost of blood product l
CD_o	Delivery cost from a blood center to a hospital using vehicle o
CH_o	Unit handling cost for delivery using vehicle o
CI_l	Unit inventory holding cost for blood product l
PS_l	Penalty cost per unit of shortage of blood product l
PO_l	Penalty cost per unit of outdated blood product l
PP	Penalty cost for disposing of excess collected blood that exceeds production capacity
PD	Penalty cost for disposing of rejected blood products
PSh	Penalty time for shortage of blood products
OC_k	Collection time per unit of blood bag type k
OT	Travel time per km from a mobile site to a blood center
OP_{jl}	Production time at blood center j per unit of product l
OI_l	Estimated inventory time per unit of blood product l
OD_o	Delivery time per km from a blood center to a hospital using vehicle o
r_{jl}	Rejection rate of blood product l at blood center j
n_{kl}	Number of product fractions obtained from blood bag k for product l
P_{kt}	Maximum available donors for blood bag type k in period t
SP_{ikt}	Available donors collected at mobile site i using blood bag k in period t
SP_{jkt}	Available donors collected at blood center j using blood bag k in period t
(p^x_{kt}, p^y_{kt})	Coordinates of supply point $p \in P_k$ in period t
(i^x_t, i^y_t)	Coordinates of mobile site $i \in I_t$ in period t
(j^x, j^y)	Coordinate of blood center j
D_{slt}	Demand at hospital s for blood product l in period t
$d(i,j,t)$	Euclidean distance from mobile site i to blood center j in period t
$d(j,s)$	Euclidean distance from blood center j to hospital s
R_{ij}	Desired radius from cluster i to blood center j
R_j	Coverage radius of blood center j
BM	Maximum mobile collection capacity
BP_{jl}	Maximum production capacity of blood center j for product l
BV_o	Maximum capacity of delivery vehicle o
E_l	Maximum shelf life of product l in days
II_{jla}	Initial inventory at blood center j of product l with age a

Table 4
Decision variables of the model.

Variable	Definition
u_{it}	Binary variable equal to 1 if mobile-site candidate i is activated in Phase I during period t
w_{ipkt}	Binary variable equal to 1 if supply point $p \in P_k$ is covered by mobile site i at period t
w_{jpkt}	Binary variable equal to 1 if supply point $p \in P_k$ is covered by center j in period t
m_{ijt}	Binary variable equal to 1 if mobile site i is assigned to blood center j and operated in period t
v_{jsot}	Binary variable equal to 1 if blood center j delivers products to hospital s using vehicle o in period t
x_{ijkt}	Number of blood bags of type k collected from mobile site i and transported to blood center j in period t
y_{jklt}	Quantity of blood product l produced at blood center j from collection bag type k in period t
ρ_{jkt}	Number of excess blood bags of type k collected at blood center j that are not processed in period t
g_{jslaot}	Quantity of blood product l with age a delivered from blood center j to hospital s using vehicle o in period t
α_{jlat}	Inventory level of blood product l with age a at blood center j at the start of period t
μ_{jlat}	Inventory level of blood product l with age a at blood center j at the end of period t
λ_{slt}	Shortage quantity of blood product l at hospital s in period t
η_{jlt}	Quantity of outdated blood product l at blood center j in period t

at mobile collection and blood center collection sites from Eqs. (3.9) and (3.10), is used to determine the optimal decisions for improving the overall performance of the blood supply chain network. This phase is guided by socio-economic objectives to ensure an effective and balanced solution across the network. The objective functions of the proposed model are formulated within a multi-objective MILP framework. The first objective function (3.12) addresses the economic objective and aims to minimize the total network cost, following the approach of Hamdan and Diabat (2019).

$$\min Z_1 = C_{\text{collection}} + C_{\text{production}} + C_{\text{inventory}} + C_{\text{distribution}}. \tag{3.12}$$

$$C_{\text{collection}} = FM \sum_{t\in T}\sum_{i\in I_t}\sum_{j\in J} m_{ijt} + \sum_{t\in T}\sum_{j\in J}\sum_{k\in K} CC_k \cdot \left[SP_{jkt} + \sum_{i\in I_t} x_{ijkt}\right] + \sum_{t\in T}\sum_{j\in J}\sum_{i\in I_t} CT \cdot d(i,j,t)\cdot m_{ijt}. \tag{3.13}$$

$$C_{\text{production}} = \sum_{t\in T}\sum_{j\in J}\sum_{k\in K}\sum_{l\in L} CP_l \cdot y_{jklt} + \sum_{t\in T}\sum_{j\in J}\sum_{k\in K}\sum_{l\in L} PD \cdot n_{kl}\cdot r_{jl}\cdot y_{jklt} + \sum_{t\in T}\sum_{j\in J}\sum_{k\in K} PP\cdot \rho_{jkt}. \tag{3.14}$$

$$C_{\text{inventory}} = \sum_{t\in T}\sum_{j\in J}\sum_{l\in L}\sum_{a\in A_l} CI_l \cdot \mu_{jlat} + \sum_{t\in T}\sum_{j\in J}\sum_{l\in L} PO_l \cdot \eta_{jlt}. \tag{3.15}$$

$$C_{\text{distribution}} = \sum_{t\in T}\sum_{j\in J}\sum_{s\in S}\sum_{o\in O} CD_o \cdot d(j,s)\cdot v_{jsot} + \sum_{t\in T}\sum_{s\in S}\sum_{l\in L} PS_l \cdot \lambda_{slt} + \sum_{t\in T}\sum_{j\in J}\sum_{s\in S}\sum_{l\in L}\sum_{a\in A_l}\sum_{o\in O} CH_o \cdot g_{jslaot}. \tag{3.16}$$

Total collection cost (3.13) consists of three parts. The first term represents the cost of establishing mobile donation sites in each period. The second term represents the blood collection cost for each blood bag type in each period. The last term represents the transportation cost from mobile donation sites to blood centers. The production cost

(3.14) also consists of three parts. The first part is the total cost of producing blood product l from blood bag type k at blood center j in each period. The second part is the disposal cost of products rejected during quality testing. The third part is the disposal cost of excess collected blood due to production capacity limits. The inventory cost component (3.15) includes two terms. The first term represents the total inventory holding cost in each period. The second term is the total disposal cost of outdated products in each period. For the distribution cost (3.16), the first term is the total delivery cost of blood products to hospitals. The second term represents the total shortage penalty cost in each period. The third term is the total handling cost of delivering blood products using different vehicle types. The second objective represents the social dimension of the model by minimizing the total time that blood products spend in the network, from collection through delivery to the hospital, which represents the point of patient demand in this study (Hosseini et al., 2023).

$$\min Z_2 = T_{\text{collection}} + T_{\text{production}} + T_{\text{inventory}} + T_{\text{distribution}}. \tag{3.17}$$

$$T_{\text{collection}} = \sum_{t\in T}\sum_{j\in J}\sum_{k\in K} OC_k \cdot \left[SP_{jkt} + \sum_{i\in I_t} x_{ijkt}\right] + \sum_{t\in T}\sum_{j\in J}\sum_{i\in I_t} OT \cdot d(i,j,t) \cdot \sum_{k\in K} x_{ijkt}. \tag{3.18}$$

$$T_{\text{production}} = \sum_{t\in T}\sum_{j\in J}\sum_{k\in K}\sum_{l\in L} OP_{jl} \cdot n_{kl} \cdot y_{jklt}. \tag{3.19}$$

$$T_{\text{inventory}} = \sum_{t\in T}\sum_{j\in J}\sum_{l\in L}\sum_{a\in A_l} OI_l \cdot \mu_{jlat}. \tag{3.20}$$

$$T_{\text{distribution}} = \sum_{t\in T}\sum_{j\in J}\sum_{s\in S}\sum_{l\in L}\sum_{a\in A_l}\sum_{o\in O} OD_o \cdot d(j,s) \cdot g_{jslaot} + \sum_{t\in T}\sum_{s\in S}\sum_{l\in L} PSh \cdot \lambda_{slt}. \tag{3.21}$$

For collection time (3.18), the first term represents the total time required for blood collection, while the second term is the transportation time from mobile donation sites to production sites. Terms (3.19)–(3.21) represent the production time, inventory time of blood products, and distribution time, including the shortage penalty time, respectively. To keep the flow-time objective dimensionally homogeneous, each transport term is weighted by the quantity actually transported, that is, by the number of bags collected in (3.18) and by the number of units delivered in (3.21), rather than by the binary routing decision; consequently, all four components of Z_2 carry units of unit·days. In the second phase, several constraints are incorporated to develop the overall network design.

$$\sum_{j\in J} m_{ijt} \le 1, \qquad \forall i \in I_t, t \in T, \tag{3.22}$$

$$\sum_{i\in I_t} m_{ijt} \le NM_j, \qquad \forall j \in J, t \in T, \tag{3.23}$$

$$x_{ijkt} \le SP_{ikt} \cdot m_{ijt}, \qquad \forall i \in I_t, j \in J, k \in K, t \in T, \tag{3.24}$$

$$\sum_{k\in K} x_{ijkt} \le BM, \qquad \forall i \in I_t, j \in J, t \in T. \tag{3.25}$$

The supply constraints are adapted from Hosseini et al. (2023). Constraint (3.22) states that each mobile site can serve at most one blood center. Constraint (3.23) specifies that the number of active mobile sites assigned to blood center j must not exceed the maximum number of mobile facilities available at that center. Constraint (3.24) states that the supply of each blood bag type k is limited by the available supply at mobile site i, and that supply can be sent from mobile site i to blood center j only when the site is active. Finally, constraint (3.25) states that the total supply collected at a mobile site is limited by the mobile facility's capacity.

$$y_{jklt} \le \left[\sum_{i\in I_t} x_{ijkt} + SP_{jkt}\right] \cdot n_{kl}, \qquad \forall j \in J, k \in K, l \in L, t \in T, \tag{3.26}$$

$$\sum_{k\in K} y_{jklt} \le BP_{jl}, \qquad \forall j \in J, l \in L, t \in T, \tag{3.27}$$

$$\rho_{jkt} = \left[\sum_{i\in I_t} x_{ijkt} + SP_{jkt}\right] - y_{jkl^*t}, \qquad \forall j \in J, k \in K, t \in T. \tag{3.28}$$

Production constraints are developed based on Abad et al. (2023). Constraint (3.26) states that the number of products produced is limited by availability. Constraint (3.27) states that production quantities are restricted by the capacity of each blood center j. Constraint (3.28) accounts for the collected blood bags that remain unprocessed at each blood center. A collected bag is consumed at most once; because every bag type is fractionated into exactly one unit of a common product, denoted by the reference product $l^* \in L$ (the product for which $n_{kl^*} = 1$ for all $k \in K$), the number of bags processed at center j in period t equals y_{jkl^*t}, and the excess ρ_{jkt} is therefore defined per collection type k rather than per product l.

$$\alpha_{jla1} = II_{jla}, \qquad \forall j \in J, l \in L, a \in A_l, \tag{3.29}$$

$$\alpha_{jl0t} = 0, \qquad \forall j \in J, l \in L, t \in T \setminus \{1\}, \tag{3.30}$$

$$\alpha_{jlat} = \mu_{jl(t-1)(a-1)}, \qquad \forall j \in J, l \in L, t \in T \setminus \{1\}, a \in A_l \setminus \{0\}, \tag{3.31}$$

$$\mu_{jl0t} = \sum_{k\in K} (1 - r_{jl}) \cdot y_{jklt}, \qquad \forall j \in J, l \in L, t \in T, \tag{3.32}$$

$$\mu_{jlat} = \alpha_{jlat} - \sum_{o\in O}\sum_{s\in S} g_{jslaot}, \qquad \forall j \in J, l \in L, a \in A_l \setminus \{0\}, t \in T, \tag{3.33}$$

$$\eta_{jl1} = 0, \qquad \forall j \in J, l \in L, \tag{3.34}$$

$$\eta_{jlt} = \mu_{jlE_l(t-1)}, \qquad \forall j \in J, t \in T \setminus \{1\}, l \in L. \tag{3.35}$$

The inventory constraints are developed based on the approach proposed by Dehghani et al. (2021). Constraint (3.29) initializes the inventory level at the beginning of period $t = 1$. Constraint (3.30) states that no product with age 0 exists in inventory at the beginning of each period. Constraint (3.31) defines the relationship between the inventory at the beginning of a period and the inventory at the end of the previous period, ensuring continuity over time. Constraint (3.32) defines the end-of-period inventory with age 0 in a given period as the total production in that period, reflecting the addition of newly produced blood products to inventory. Constraint (3.33) governs inventory with age greater than 0 by linking it to the outflow of products during the period, managing aged inventory and its utilization. Constraint (3.34) assumes that there are no outdated products in the first period, simplifying model initialization. Finally, Constraint (3.35) accounts for the quantity of outdated products in each period, representing items that exceed their maximum shelf lives.

$$\sum_{o\in O} v_{jsot} \le 1, \qquad \forall j \in J, s \in S, t \in T, \tag{3.36}$$

$$\sum_{l\in L}\sum_{a\in A_l} g_{jslaot} \le BV_o \cdot v_{jsot}, \qquad \forall j \in J, s \in S, o \in O, t \in T, \tag{3.37}$$

$$\sum_{s\in S}\sum_{o\in O} g_{jslaot} \le \alpha_{jlat}, \qquad \forall j \in J, l \in L, a \in A_l, t \in T, \tag{3.38}$$

$$\sum_{a\in A_l}\sum_{o\in O}\sum_{j\in J} g_{jslaot} \le D_{slt}, \qquad \forall s \in S, l \in L, t \in T, \tag{3.39}$$

$$\lambda_{slt} = D_{slt} - \sum_{a\in A_l}\sum_{o\in O}\sum_{j\in J} g_{jslaot}, \qquad \forall s \in S, l \in L, t \in T. \tag{3.40}$$

Constraint (3.36) states that only one type of vehicle can be used for delivery from blood center j to hospital s. Constraint (3.37) ensures that the quantity transported does not exceed the maximum capacity of the selected vehicle, and the product flow can occur only if vehicle type o is selected for delivery from blood center j to hospital s (Hosseini et al., 2023). Constraint (3.38) states that the quantity of transported product is limited by the inventory level at the beginning of the period (Jabbarzadeh et al., 2014). Constraint (3.39) defines demand

$$u_{it}, w_{ipkt}, w_{jpkt}, m_{ijt}, v_{jsot} \in \{0,1\}, \quad \forall i \in I_t, j \in J, p \in P_k, k \in K, t \in T, \tag{3.41}$$

$$x_{ijkt}, y_{jklt}, \alpha_{jlat}, \mu_{jlat}, g_{jslaot}, \lambda_{slt}, \eta_{jlt}, \rho_{jkt} \geq 0, \quad \forall i \in I_t, j \in J, s \in S, k \in K, l \in L, a \in A_l, t \in T. \tag{3.42}$$

Box I.

fulfillment, while constraint (3.40) determines the shortage in each period (see Eqs. (3.41) and (3.42) in Box I).

Finally, constraints (3.41) and (3.42) define the binary and non-negativity restrictions for the decision variables.

4. Solution methodology

This section presents the solution methodology for the problem described in the previous subsection. The proposed approach consists of two phases. In Phase I, k-means clustering is used to identify feasible locations for mobile donation sites in each period. In Phase II, the multi-objective MILP model is solved utilizing the ϵ-constraint method.

4.1. Phase I: k-means clustering

In the first phase, the focus is on determining suitable locations for mobile donation sites using a clustering method for each period. Among the available clustering methods, k-means clustering is employed due to its simplicity and effectiveness in handling spatial clustering problems (Gupta et al., 2021). In this study, the centroids obtained from the clustering process are selected as the locations of mobile donation sites. The clustering model follows the formulation presented in the previous subsection and uses Eqs. (3.2)–(3.8) as feasibility constraints.

The two-phase separation prevents the fractional centroid expressions in Eqs. (3.6)–(3.7) from being embedded directly in the multi-period operational model, thereby preserving the MILP structure of Phase II. This decomposition restricts Phase II to the candidate locations generated in Phase I; consequently, the comparison of alternative clustering methods in Section 5.5 evaluates the sensitivity of the operational results to candidate generation.

4.2. Phase II: ϵ-constraint method

In the second phase, the multi-objective socio-economic optimization problem is solved using the ϵ-constraint method (Mavrotas, 2009), which converts the problem into a single-objective model. This phase builds upon the results obtained from Phase I and addresses the objective functions in Eqs. (3.12) and (3.17), subject to the constraints in Eqs. (3.22)–(3.42). In the ϵ-constraint method, one objective function is selected as the primary objective, while the other objective is treated as a constraint bounded by a specific ϵ value. First, a single-objective optimization is performed using Eq. (3.12), denoted by Z_1, along with constraints (3.22)–(3.42). Let β_{Z_1} denote the decision variable vector obtained from this process. Then, $Z_1(\beta_{Z_1})$ gives the lower bound of Z_1, designated as LB_{Z_1}, while $Z_2(\beta_{Z_1})$ calculates the upper bound of Z_2, denoted by UB_{Z_2}. Next, a single-objective optimization is performed using Eq. (3.17), denoted by Z_2, together with constraints (3.22)–(3.42). Let β_{Z_2} represent the decision variable vector obtained from this optimization. In this case, $Z_2(\beta_{Z_2}) = LB_{Z_2}$ and $Z_1(\beta_{Z_2}) = UB_{Z_1}$.

After obtaining the upper and lower bounds for both objective functions, the multi-objective problem in the second phase, with the objective function $\min(Z_1, Z_2)$ and constraints (3.22)–(3.42), is reformulated into two ϵ-constraint problem models.

In the first model, Eq. (3.12) is used as the objective function, subject to constraints defined by Eqs. (3.22)–(3.42), while Eq. (3.17) is reformulated as an additional constraint (4.43) with the ϵ value bounds defined in Eq. (4.44).

$$Z_2 \leq \epsilon_{Z_2}. \tag{4.43}$$

$$LB_{Z_2} \leq \epsilon_{Z_2} \leq UB_{Z_2}. \tag{4.44}$$

In the second model, Eq. (3.17) is used as the objective function, subject to constraints defined by Eqs. (3.22)–(3.42), while Eq. (3.12) is reformulated as an additional constraint (4.45) with the ϵ value bounds defined in Eq. (4.46).

$$Z_1 \leq \epsilon_{Z_1}, \tag{4.45}$$

$$LB_{Z_1} \leq \epsilon_{Z_1} \leq UB_{Z_1}. \tag{4.46}$$

By solving both ϵ-constraint models, the trade-off between the two objective functions can be analyzed, and the ϵ turning point can be identified where the two objectives achieve balance.

Each ϵ-constrained model is solved for a sequence of ϵ values spanning the interval $[LB, UB]$ of the bounded objective, so that each value yields one non-dominated solution and the collection of solutions traces the Pareto front. To select a single most preferred solution from this front, the two objectives are first placed on a common scale by normalizing each against its bounds, $\delta_1 = (Z_1 - LB_{Z_1})/(UB_{Z_1} - LB_{Z_1})$ and $\delta_2 = (Z_2 - LB_{Z_2})/(UB_{Z_2} - LB_{Z_2})$. The preferred (turning) solution is the generated non-dominated solution that minimizes the larger of the two normalized objectives,

$$\min \max\{\delta_1, \delta_2\}.$$

This criterion selects the point closest to the ideal point (LB_{Z_1}, LB_{Z_2}) in the worst-case normalized sense, that is, the best-balanced solution in which neither the economic nor the social objective is disproportionately compromised.

5. Case study

To evaluate the performance of the proposed model, two numerical case studies are conducted using real-world data from the BSC network in Indonesia. In the first phase, the k-means clustering package is utilized, while OSMNX is employed to obtain and visualize the geographic locations of nodes on the instance maps. Inter-node distances used in the model are computed as Euclidean distances from these coordinates (Assumption 5); given the compact regional extent of the network, in which all nodes lie within roughly a 30 km radius, the Euclidean approximation is adequate for the strategic planning addressed here. The second phase is performed using the AMPL IDE with the Gurobi solver. All numerical experiments are executed on a laptop with an Intel Core i7-9750H CPU operating at 2.60 GHz, 12 processors, and 8 GB of RAM.

5.1. Instance generation

Given $j \in \{1,2,3\}$, case studies involve three blood centers. Case 1 includes 55 hospital locations within the observed district, while Case 2 includes 118 hospitals. This demonstrates the capability of blood centers to serve hospitals beyond their designated responsibilities and better reflects the actual conditions of the BSC network.

Table 5
Cost components.

Cost component		Price (IDR)
CP_l	FM	317,800
	CT	1000
	CC_k	50,000
	PRC	175,000
	TC	175,000
CI_l	PRC	2000
	TC	15,000
CD_o	Motorcycle	200
	Van	1000
CH_o	Motorcycle	200
	Van	100

Table 6
Time components.

Time component			Time (Days)
Collection	Type 1		0.0083
	Type 2		0.0333
Transportation to BC			0.0026
Production	BC1	PRC	0.0109
		TC	0.034
	BC2	PRC	0.0513
		TC	0.1
	BC3	PRC	0.0268
		TC	0.05
Inventory	PRC		1
	TC		1
Distribution to Hospital	Motorcycle		0.0018
	Van		0.0026

Table 7
Product conversion rates.

Process	PRC	TC
Double bag	1	0
Triple bag	1	1

5.2. Parameter settings

Using different time frames of data, Cases 1 and 2 are constructed from demand and supply data obtained from three blood centers in the special region of Yogyakarta, Indonesia, namely BC1 (Kota Yogyakarta), BC2 (Bantul), and BC3 (Sleman). Case 1 utilizes daily demand and supply data over a 60-period time horizon, covering January to February 2021, while Case 2 is based on data from November to December 2021.

The parameter settings are identical for both cases, and the cost components are adapted from Fauzi et al. (2019) and further adjusted to align with the specific problem requirements. The cost and time settings used in experiments are provided in Tables 5 and 6, respectively. The blood bag types considered are double and triple blood bags, designated as Type 1 and Type 2 collection blood bags, respectively. The final products analyzed are Packed Red Cells (PRC) and Thrombocyte Concentrate (TC), corresponding to red blood cells and platelets, respectively. Since every collection bag yields exactly one PRC unit, the reference product l^* defined in Section 3 corresponds to PRC in these experiments. The conversion rates for these products are presented in Table 7. The shelf life of PRC is 35 days, and that of TC is 5 days. The maximum production capacities for PRC and TC at each BC are given in Table 8. The maximum capacity is set at 10 units for motorcycles and 50 units for vans.

Table 8
Maximum production capacity.

Blood Center		Capacity
BC1	PRC	200
	TC	100
BC2	PRC	60
	TC	10
BC3	PRC	120
	TC	20

For each period, the clustering process in Phase I is performed based on the available mobile fleet at each blood center j, i.e., NM_j. Accordingly, the number of clusters is linked directly to the total mobile-fleet availability, such that $k = \sum_{j \in J} NM_j$. With $NM_j = 2$ collection vehicles available at each of the three blood centers, $k = 6$ candidate mobile sites are generated, whereas the $k = 3$ scenarios correspond to a reduced fleet of one vehicle per center. These two values represent the considered operational range of mobile-collection capacity and link number of candidate sites to physical vehicle availability rather than to an arbitrary choice. To examine the impact of varying the number of clusters and the spatial distribution of supply points, Case 1 is extended to four additional scenarios, as summarized in Table 9. These scenarios consider alternative supply-point distributions to reflect differences in the demographic characteristics of Indonesia. An illustrative example of the clustering results for Case 1 is depicted in Fig. 3.

To address inefficiencies, the model incorporates a penalty cost of IDR 375,000 for each outdated product and IDR 410,000 for each unit of unmet demand. The penalty cost for disposing of products due to production capacity constraints is set at IDR 100,000 per unit, and the penalty cost for disposing of rejected products is also set at IDR 100,000 per unit. The rejection rate is assumed to be zero for each BC, as there is insufficient information regarding finished products for one of the BCs. This assumption ensures consistency in the model while acknowledging the limitation of the available data. The shortage time penalty is assumed to be 5 days because the maximum time from collecting one unit to distributing it is 4 days, and the maximum shelf life of TC is 5 days.

It is assumed that, in period 1, each BC already holds an initial inventory of each product that can be used to meet demand during that period. This assumption establishes a baseline for the system by ensuring that the model begins with an existing stock rather than from zero.

5.3. Results and discussion

Phase I is applied separately to each Case 1 scenario and to Case 2 with $k = 6$. For each planning period, the cluster centroids obtained from Eqs. (3.1)–(3.11) define the candidate mobile donation sites supplied to Phase II.

For each instance, Phase II is first solved twice: once by minimizing the economic objective Z_1 and once by minimizing the responsiveness objective Z_2. Minimizing Z_1 determines LB_{Z_1} and the corresponding UB_{Z_2}, whereas minimizing Z_2 determines LB_{Z_2} and the corresponding UB_{Z_1}. Table 10 reports these bounds, which define the ϵ intervals used in the two ϵ-constraint formulations described in Section 4.2.

Table 11 presents the objective-value pairs obtained from the two single-objective runs for Case 1, Scenario 1 and Case 2. To generate the trade-off curves, Z_1 is retained as the primary objective while the bound ϵ_{Z_2} is varied uniformly from LB_{Z_2} to UB_{Z_2}. Thirteen ϵ_{Z_2} values are used for each Case 1 scenario and eleven values for Case 2. One Phase-II model is solved at each value, and the resulting trade-off solutions are plotted in Figs. 4 and 5. Because the magnitudes of Z_1 ($\sim 10^9$ IDR) and Z_2 ($\sim 10^3$ unit-days) differ substantially, the absolute MIP optimality tolerance is specified according to the primary objective: 2×10^4 IDR when minimizing Z_1 and 5×10^{-3} unit-days when minimizing

Table 9
Scenario descriptions for Case 1.

Scenario	Description
1	Random distribution of supply points, with $k = 6$ in the k-means clustering phase
2	Random distribution of supply points, with $k = 3$ in the clustering phase
3	Distribution adjusted using demographic data as weights, with $k = 6$ in the clustering
4	Distribution adjusted using demographic data as weights, with $k = 3$ in the clustering phase

Table 10
Upper and lower bounds of objective functions.

Case	Scenario	Total cost		Total time	
		Upper bound	Lower bound	Upper bound	Lower bound
1	1	1,355,434,151	1,349,408,434	3259.81	3239.74
	2	1,355,895,111	1,351,060,900	3259.92	3238.54
	3	1,362,829,035	1,355,560,546	3260.99	3240.20
	4	1,373,603,989	1,355,564,854	3261.69	3238.36
2	–	1,502,658,059	1,440,192,870	4773.31	4527.06

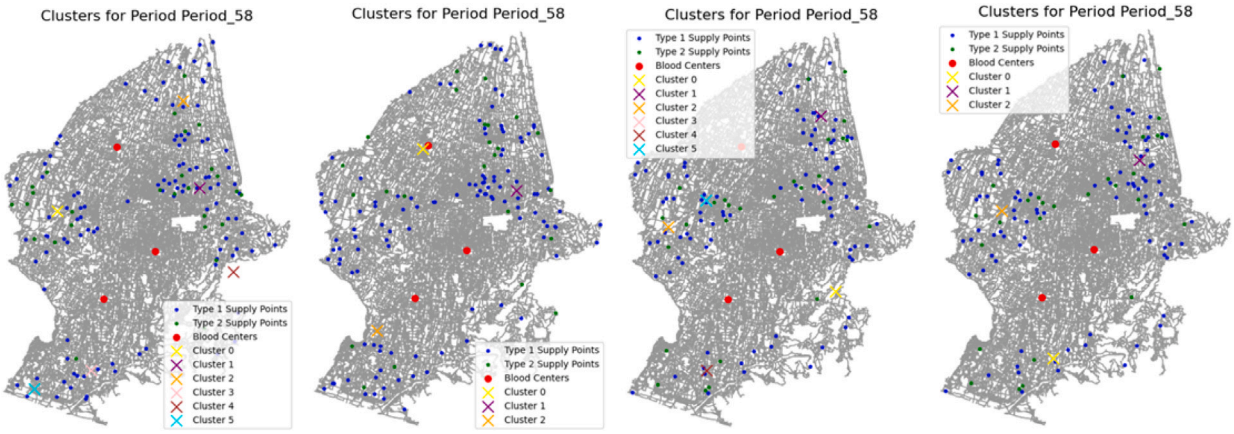


Fig. 3. Clustering example for Case 1 under Scenarios 1–4, shown from left to right.

Table 11
Payoff table for representative instances (Case 1, Scenario 1 and Case 2).

Instance	Optimization	Total cost (Z_1)	Total time (Z_2)
Case 1 (Scenario 1)	min Z_1	1,349,408,434	3259.81
	min Z_2	1,355,434,151	3239.74
Case 2	min Z_1	1,440,192,870	4773.31
	min Z_2	1,502,658,059	4527.06

Z_2. These are solver stopping tolerances and are separate from the normalized deviations used to select the turning-point solution.

Figs. 4 and 5 show differences in slope between Cases 1 and 2. Case 2 exhibits a less gradual slope, better representing the trade-off between the two objectives due to the use of more representative demand data. Specifically, total cost varies by roughly 2.5% along the Case 2 front, against less than 1% for Case 1, reflecting the tighter supply–demand balance of the larger instance. In general, both figures illustrate that a reduction in total flow time is associated with an increase in total cost. Additional experiments are also conducted for the second ϵ-constraint model, and the turning points for both cases are provided in Table 12.

Based on Table 12, the ϵ points in each numerical experiment indicate the balance between the two objectives. Notably, the two ϵ-constraint formulations converge to essentially the same turning point in every scenario, confirming the consistency of the trade-off analysis. Across the four Case-1 scenarios, the turning-point total cost varies only marginally (by less than 1%): the number of clusters, set by mobile-fleet availability, has a negligible effect on total cost, because the network is supply-abundant under normal operation, so the number and placement of mobile sites act as a coverage and responsiveness lever rather than a cost lever. The larger differences appear in Case 2, which involves a greater number of hospitals and a tighter supply–demand balance. For each turning point in the ϵ-constraint model, if the economic objective (Z_1) is prioritized, the constraint should be set to a value smaller than ε_{Z_1}. Conversely, if the social objective (Z_2) is prioritized, the constraint should be set to a value larger than ε_{Z_1}. The same principle applies to the ϵ-constraint model for Z_2, with the settings reversed according to the priority of the objectives.

5.4. Value of dynamic mobile-site selection

To isolate the contribution of dynamic, period-varying site selection, we compare the proposed design against two baselines: a design with no mobile sites, in which collection occurs only at the fixed blood centers; and a static design, in which the period-1 site selection is fixed across the entire horizon. Table 13 reports the minimum cost and minimum flow time attained under each design, together with the number of site-openings used at the time optimum.

The minimum cost is essentially identical across the three designs (within 0.001%), confirming that mobile collection is not a cost-reduction mechanism in this supply-abundant network. On responsiveness, however, the dynamic design attains the flow-time optimum by opening only 8 sites over the horizon, whereas the static design requires 180 site-openings to reach comparable responsiveness, and the no-mobile design is 7.5 flow-time units worse. Period-varying siting therefore delivers the full responsiveness benefit at a small fraction of the infrastructure commitment, which is the central value of the proposed two-phase design.

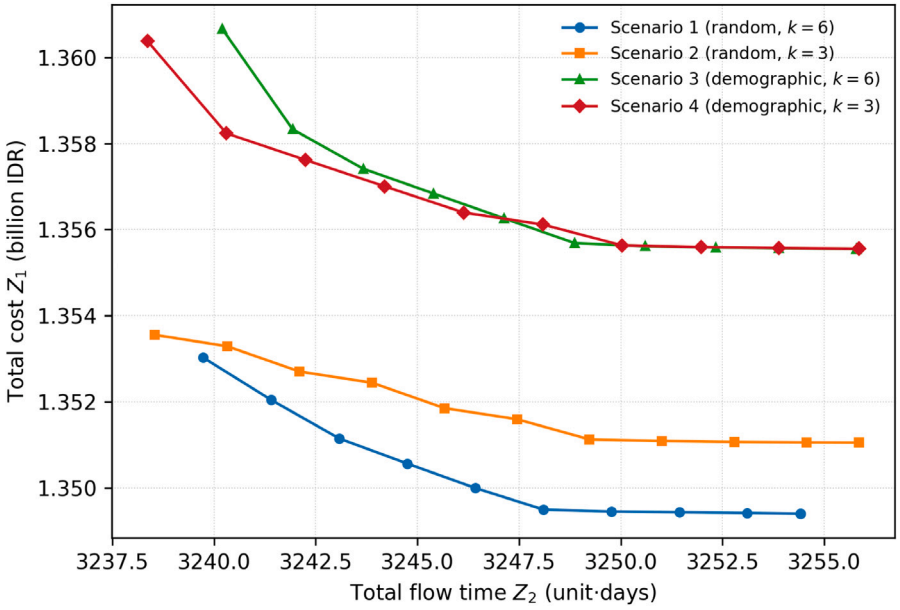


Fig. 4. Trade-off curves for Case 1 using the ϵ-constraint on Z_2 across scenarios.

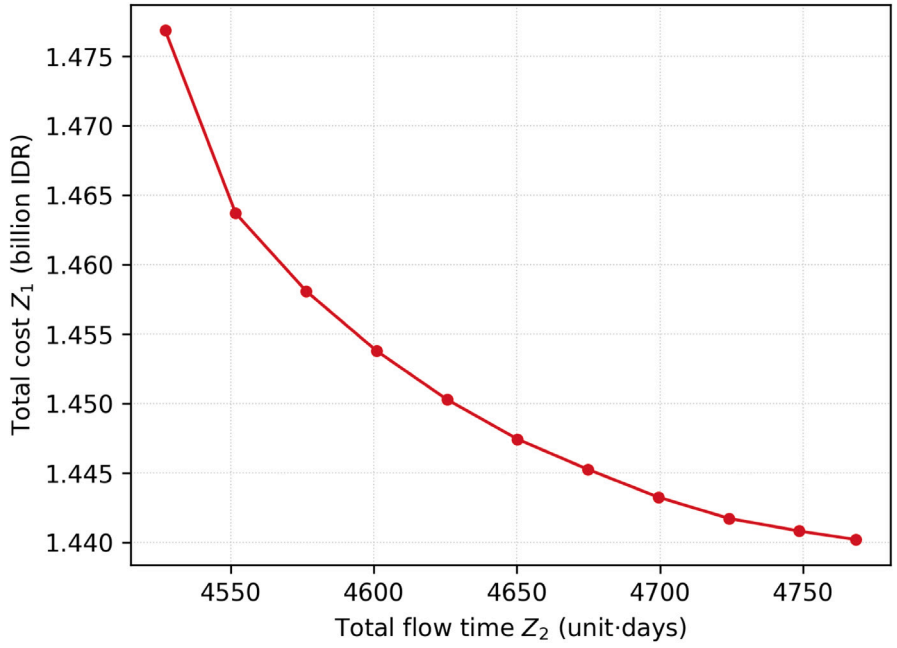


Fig. 5. Pareto front for Case 2.

Table 12
Turning points obtained from the ϵ-constraint models.

Case	Scenario	ϵ	Turning point	Total cost	Total time
1	1	Z_1	1,350,563,913	1,350,563,913	3244.75
		Z_2	3243.50	1,350,786,237	3243.50
	2	Z_1	1,352,442,981	1,352,442,981	3243.88
		Z_2	3244.13	1,352,264,422	3244.13
	3	Z_1	1,356,839,760	1,356,839,760	3245.40
		Z_2	3243.94	1,357,377,668	3243.94
	4	Z_1	1,358,233,796	1,358,233,796	3240.31
		Z_2	3240.19	1,358,571,376	3240.19
2	-	Z_1	1,455,637,399	1,455,637,399	4588.62
		Z_2	4587.53	1,455,809,167	4587.53

Table 13
Comparison of the proposed dynamic siting against fixed-location and no-mobile baselines (Case 1, Scenario 1).

Design	Min cost (IDR)	Min flow time	Site-openings at time optimum
Proposed (dynamic, period-varying)	1,349,408,434	3239.72	8
Static sites (period-1 fixed)	1,349,406,114	3239.80	180
No mobile sites	1,349,406,427	3247.22	0

Table 14
End-to-end comparison of Phase-I clustering methods (Case 1, Scenario 1, $k = 6$).

Method	Inertia	Mean centroid–BC (km)	Min cost (IDR)	Min flow time
k-means	143,353	8.49	1,349,407,523	3239.58
k-medoids	183,939	8.49	1,349,404,198	3239.74
Ward	161,534	8.36	1,349,407,385	3239.72
GMM	184,101	8.26	1,349,406,448	3239.62

Table 15
Sensitivity analysis under increasing demand (Case 1, Scenario 1).

Demand	Economic perspective			Social perspective		
Increase	Total cost	Total time	Waste cost	Total cost	Total time	Waste cost
+ 100%	1.10%	3.66%	−7.49%	3.79%	3.00%	−7.16%
+ 200%	14.56%	71.43%	−37.37%	21.58%	59.02%	−33.24%
+ 300%	30.76%	203.32%	−66.92%	46.66%	130.62%	−52.69%
+ 400%	56.43%	350.56%	−85.71%	71.79%	236.95%	−69.83%

5.5. Comparison of clustering methods

To justify the use of k-means in Phase I, we compare it against three alternative clustering methods, k-medoids (PAM) (Kaufman & Rousseeuw, 1990), agglomerative hierarchical clustering (Ward linkage) (Ward Jr, 1963), and Gaussian mixture models (GMM) (McLachlan & Peel, 2000), by running each through the complete Phase II model on an identical supply geography. Table 14 reports, for each method, the within-cluster dispersion (inertia) and the resulting Phase II economic and responsiveness objectives.

Two observations follow. First, k-means yields the most compact clusters (lowest inertia), the property most relevant to siting collection points close to donors. Second, and more importantly, the end-to-end objectives are nearly invariant to the clustering method: total cost varies by less than 0.001% and total flow time by less than 0.01% across all four methods. The two-phase framework is therefore robust to the Phase-I clustering choice, and k-means is justified as a simple, compact, and effective option. This near-invariance also indicates that the Phase-I restriction is not the binding factor in the achievable Phase-II objective, supporting the quality of the two-phase decomposition. Because the original Phase-I point set was not retained, this comparison uses a supply geography reconstructed to match the recorded per-period donor counts and the region's spatial extent; it is thus a method-relative comparison on a representative instance.

6. Sensitivity analysis

In both numerical cases, the total supply during the observation period is sufficient to satisfy all demands, as evidenced by the absence of shortages across all scenarios. To further evaluate the responsiveness of the model and its behavior under adverse conditions, sensitivity analyses are performed. The first sensitivity analysis considers a scenario of increasing demand based on Case 1, Scenario 1. In this scenario, the overall product demand in each period is gradually increased over a 60-period horizon. The analysis is conducted using both ϵ-constraint formulations, and the waste cost is then calculated using Eq. (6.47).

$$\text{Waste Cost} = \sum_{t\in T}\sum_{j\in J}\sum_{k\in K} PP\cdot \rho_{jkt} + \sum_{t\in T}\sum_{j\in J}\sum_{l\in L} PO_l\cdot \eta_{jlt} + \sum_{t\in T}\sum_{j\in J}\sum_{k\in K}\sum_{l\in L} PD\cdot n_{kl}\cdot r_{jl}\cdot y_{jklt} \tag{6.47}$$

Table 16
Sensitivity analysis under varying laboratory rejection rate (Case 1, Scenario 1).

Rejection rate r_{jl}	Total cost (IDR)	Total time	Waste cost (IDR)
0.00	1,349,408,434	3259.81	613,200,000
0.10	1,391,825,824	3269.78	617,455,556
0.20	1,444,866,323	3286.19	622,775,000
0.30	1,513,059,401	3308.27	629,614,286

Table 15 shows a consistent increase in total cost and time as demand increases, from both social and economic perspectives. Additionally, the analysis reveals a decrease in waste cost as demand rises. This occurs because more collected blood is allocated to production to satisfy the higher demand. The scenario is extended until demand increases by 400%, at which point a shortage of 17 thrombocyte units is observed under the economic objective. It can be concluded that the optimization method effectively identifies decisions to avoid shortages by increasing supply collection and opening more mobile donation sites while minimizing total waste in terms of cost. Fig. 6 plots these changes for both objectives.

We further examine the effect of the laboratory rejection rate r_{jl}, which is set to zero in the base case owing to insufficient finished-product test data for one blood center. Table 16 reports total cost, flow time, and waste as r_{jl} is varied uniformly from 0 to 0.30, with the waste cost computed from the corrected expression (6.47), which now includes the rejection-disposal term.

As the rejection rate rises to 0.30, total cost increases by about 12% and waste rises monotonically, while no shortages occur at any rejection level. The zero-rejection assumption is therefore conservative with respect to service level and does not affect the qualitative conclusions.

To examine the production echelon, we next assess whether expanding production capacity at the less-equipped centers (BC 2 and BC 3) improves network performance. Given that BC 1 is equipped with the most advanced facilities, the analysis focuses on the two smaller centers. Table 17 reports the utilization of production capacity at the cost optimum, expressed as the mean and maximum quantity produced per period against the installed capacity, together with the number of periods in which capacity is reached.

As Table 17 shows, production capacity is essentially non-binding: mean utilization is well below the installed capacity at every center

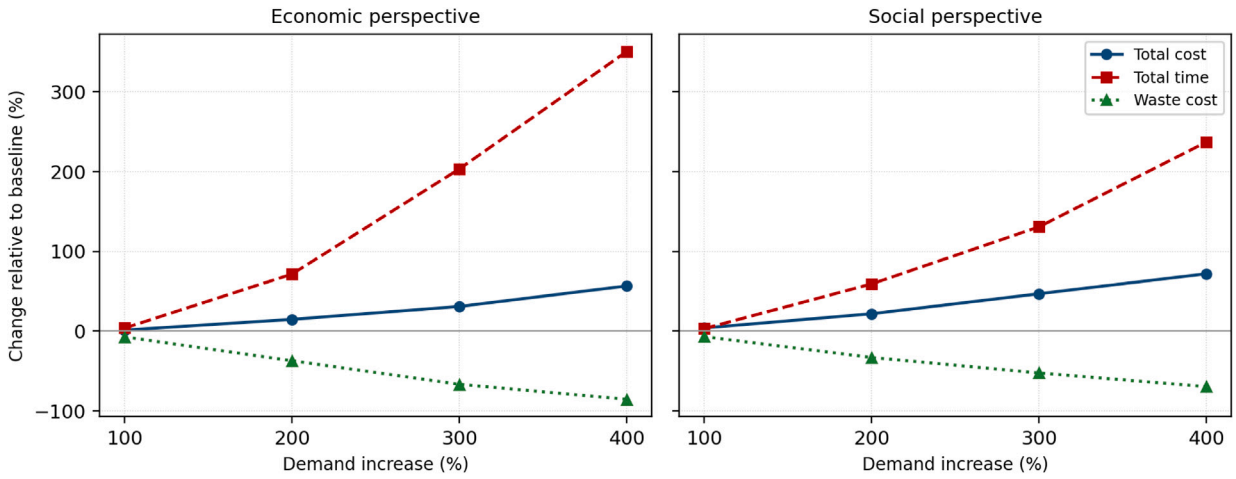


Fig. 6. Sensitivity analysis under increasing demand (Case 1, Scenario 1): percentage change in total cost, total flow time, and waste cost relative to the baseline, under the economic and social objectives.

Table 17
Production capacity utilization at the cost optimum (Case 1, Scenario 1).

Blood center	Product	Capacity	Mean used	Max used	Periods at capacity
BC1	PRC	200	17.2	65	0
	TC	100	4.2	14	0
BC2	PRC	60	4.9	23	0
	TC	10	0.9	10	1
BC3	PRC	120	4.2	25	0
	TC	20	1.3	10	0

Table 18
Sensitivity analysis under decreasing supply (Case 2).

Supply	Economic perspective			Social perspective		
decrease	Total cost	Total time	Waste cost	Total cost	Total time	Waste cost
Normal	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
−10%	−9.00%	4.21%	−16.83%	−7.54%	0.51%	−14.92%
−20%	−16.01%	9.23%	−30.08%	−15.03%	0.79%	−26.68%
−30%	−22.96%	16.70%	−43.88%	−19.74%	1.50%	−38.78%
−40%	−29.27%	19.17%	−56.60%	−25.45%	2.41%	−50.47%
−50%	−34.96%	23.83%	−68.95%	−30.32%	3.47%	−60.44%

and product, and capacity is reached in only one of the 360 center–product–period combinations. Consequently, expanding the capacity of BC 2 or BC 3 by up to a factor of 3.5 leaves the total network cost unchanged (variation below 0.001%). Even under a 400% demand increase, the residual thrombocyte shortage reported above is not relieved by additional capacity, because it is limited by the availability of triple collection bags, the sole source of thrombocytes, rather than by processing capacity. We therefore conclude that raw-material supply, not processing capacity, is the binding resource in this network, which motivates the supply-side resilience analysis that follows.

In Case 2, the scenario gradually decreases the overall product supply in each period over a 60-period horizon. The sensitivity analysis is conducted using both ϵ-constraint formulations and the waste cost (6.47). This analysis is performed because no shortage is observed under the baseline condition. However, at the turning point, the model introduces the decision to open mobile donation sites, which is not observed in Case 1. Since Case 2 provides a more representative demand setting due to the larger number of hospitals, additional scenarios are examined under reduced blood center supply conditions.

Under the reduced-supply scenario, the two objectives diverge in a policy-relevant way, as shown in Table 18. Minimizing cost, the network begins to accept shortages once supply falls by 30% (21, 63, and 149 unmet units at the 30%, 40%, and 50% reductions, respectively), because paying the shortage penalty becomes cheaper than additional collection. Minimizing flow time, the network instead activates mobile donation sites and avoids shortages entirely down to a 50% supply reduction. This contrast quantifies the responsiveness value of mobile collection under scarcity and is consistent with the dynamic-siting comparison discussed earlier. The corresponding changes in cost, time, and waste are plotted in Fig. 7. In Case 2, the changes in cost, time, and waste become more pronounced as the supply decreases.

7. Conclusion and future scope

The proposed BSC network design employs a two-phase optimization framework. Phase I uses k-means clustering to identify potential mobile donation sites dynamically in each period, whereas Phase II formulates the multi-period, multi-product operational decisions as a multi-objective MILP. The Phase-II model incorporates vehicle heterogeneity and product-specific perishability, and the ϵ-constraint method is used to analyze the economic–responsiveness trade-off and identify a balanced solution.

Two numerical case studies are utilized to evaluate the model's performance. Case 1 is divided into four scenarios based on the distribution of supply points and the number of clusters representing possible mobile donation site locations. The results indicate that the value of the two-phase design lies in responsiveness rather than cost: dynamic, period-varying mobile siting attains the responsiveness of a

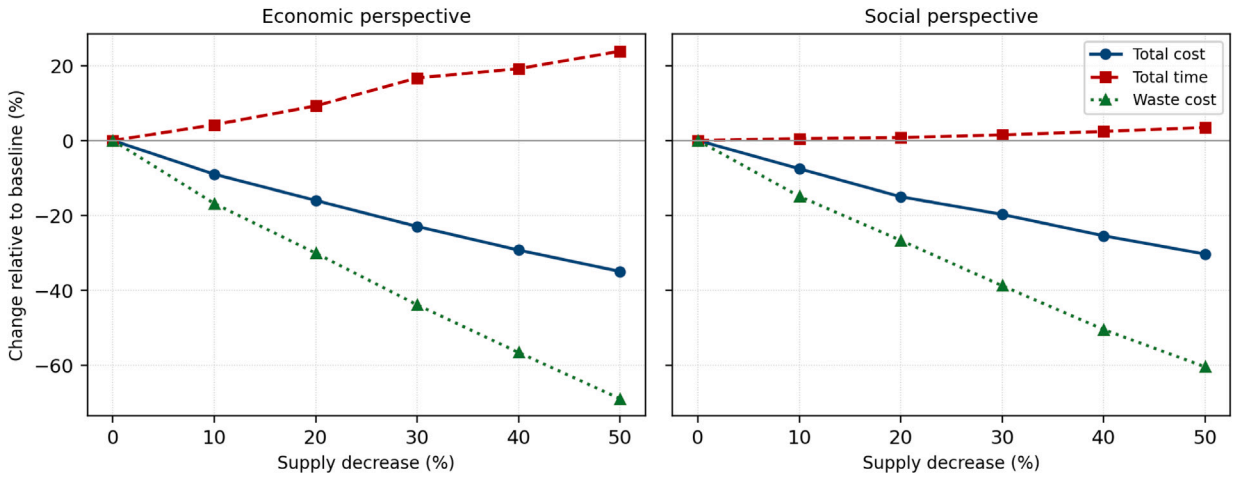


Fig. 7. Sensitivity analysis under decreasing supply (Case 2): percentage change in total cost, total flow time, and waste cost relative to the normal condition, under the economic and social objectives.

static, fixed-location design while opening substantially fewer sites, and it sustains service under substantial supply reduction. The number of clusters, which is set by mobile-fleet availability, has a negligible effect on total cost, so it acts as a coverage and responsiveness lever rather than a cost lever. A comparison of clustering methods further shows that the end-to-end objectives are essentially insensitive to the Phase-I clustering algorithm, confirming that the contribution rests on the two-phase structure and dynamic siting rather than on a particular clustering method. Through scenario development and sensitivity analyses on increasing demand and decreasing supply, the model consistently identifies decisions that minimize shortages while reducing waste in terms of cost. A network analysis further reveals that, in the studied network, processing capacity is not the binding resource; raw-material supply and demand are, which we recommend as the focus of operational planning.

To improve the model further, future research could incorporate additional factors into the k-means clustering process beyond distance, and explore capacity- and accessibility-aware clustering to improve location selection. Future research should also benchmark the proposed two-phase framework against a fully integrated formulation on suitably designed small-scale instances for which complete Phase-I spatial data are available, thereby allowing the solution-quality gap introduced by the decomposition to be quantified explicitly. Extending the deterministic formulation to explicitly capture demand and supply uncertainty through stochastic or robust optimization, for instance, a distributionally-robust supply variant, is an important direction for future work that would increase the model's applicability. In addition, optimizing different blood bag collection types and their selection would further refine its effectiveness. Integrating transshipment between blood centers could maximize utilization and reduce waste, and addressing product returns from hospitals would add a valuable dimension to the network design. Adopting road-network distances in place of the Euclidean approximation would refine travel-time estimates. Finally, incorporating blood types and Rhesus factors (ABO/Rh) would improve the model's practical relevance by reflecting the complexities of real-world blood supply chains. These enhancements would collectively extend the model's applicability to the challenges of modern blood supply chains.

CRediT authorship contribution statement

Vincent F. Yu: Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Annisa Cipta Nabila:** Writing – original draft, Validation, Software, Methodology, Data curation. **Pham Tuan Anh:** Writing – review & editing, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Soumen Kumar Das:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Achmad Pratama Rifai:** Writing – review & editing, Visualization, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data supporting this study are provided in aggregate form (daily counts of donations, blood issued, and inventory by product and blood group). Individual-level donor and patient records cannot be shared because they contain personal and health data protected under applicable data-protection regulations. The aggregated data are available from the corresponding author on reasonable request.

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